

NLP

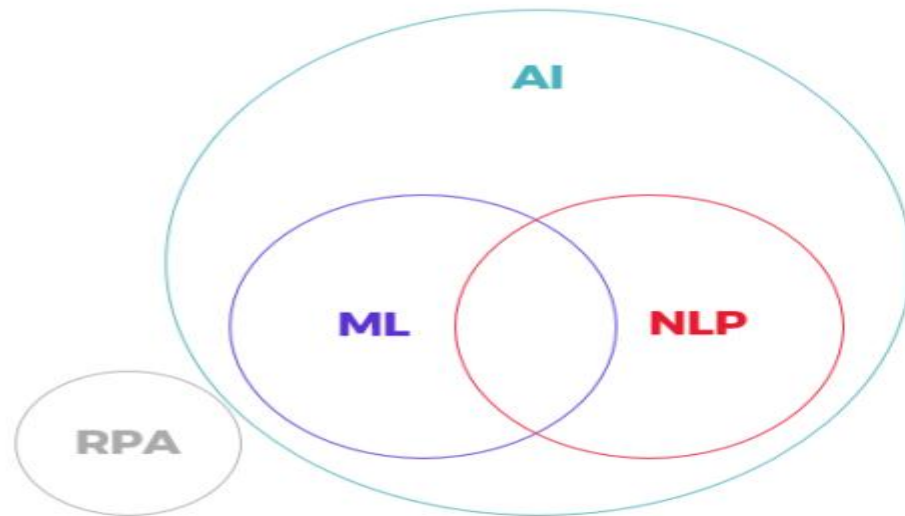
Computational Linguistics

Information engineering

Software engineering

Artificial intelligence

AI IN PROCUREMENT - DEFINITIONS



- **Artificial Intelligence (AI):** algorithms exhibiting any behavior considered 'smart'.
- **Machine Learning (ML):** algorithms that detect patterns and use them for prediction or decision making.
- **Natural Language Processing (NLP):** algorithms which can interpret, transform and generate human language.
- **Robotic Process Automation (RPA):** algorithms that mimic human actions to reduce repetitive, simple tasks.

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Data



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Information



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Knowledge



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Intelligence

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- Language is one of the oldest creations of mankind for enhancing science, business and useful research. With the advent of computer in all walks of life in the modern world, there is a need of a technology which can make direct interpretation of natural language with the help of computers. This requires the machine to be intelligent.

- Natural language processing, technically called as Computational Linguistic aims at interpretation of the natural language text by machine at a level comparable with human. Computers are nowadays used for acquisition, transmission, monitoring, storage, analysis, and transformation of information Hence, endowing them with the ability to understand and generate information expressed in natural language is a major field of research in Artificial Intelligence

- Natural Language Processing (NLP) is the computerized approach to analyzing text that is based on both a set of theories and a set of technologies. And, being a very active area of research and development, there is not a single agreed-upon definition that would satisfy everyone, but there are some aspects, which would be part of any knowledgeable person's definition. The definition I offer is: Definition: Natural Language Processing is a theoretically motivated range of computational techniques for analyzing and representing naturally occurring texts

- at one or more levels of linguistic analysis for the purpose of achieving human-like language processing for a range of tasks or applications. Several elements of this definition can be further detailed. Firstly the imprecise notion of

- 'Range of computational techniques' is necessary because there are multiple methods or techniques from which to choose to accomplish a particular type of language analysis.

- 'Naturally occurring texts' can be of any language, mode, genre, etc. The texts can be oral or written. The only requirement is that they be in a language used by humans to

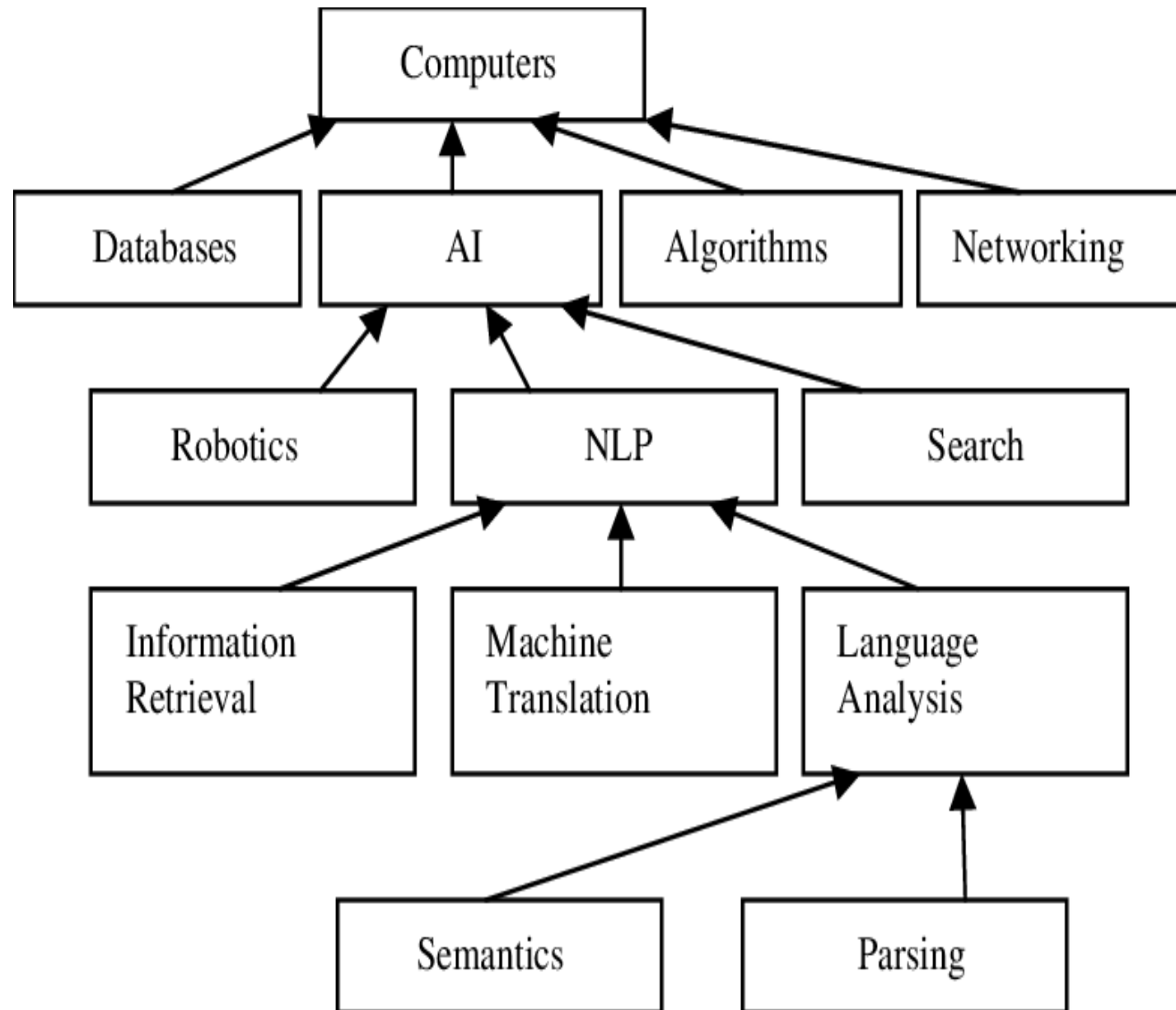
- Communicate to one another

History of NLP

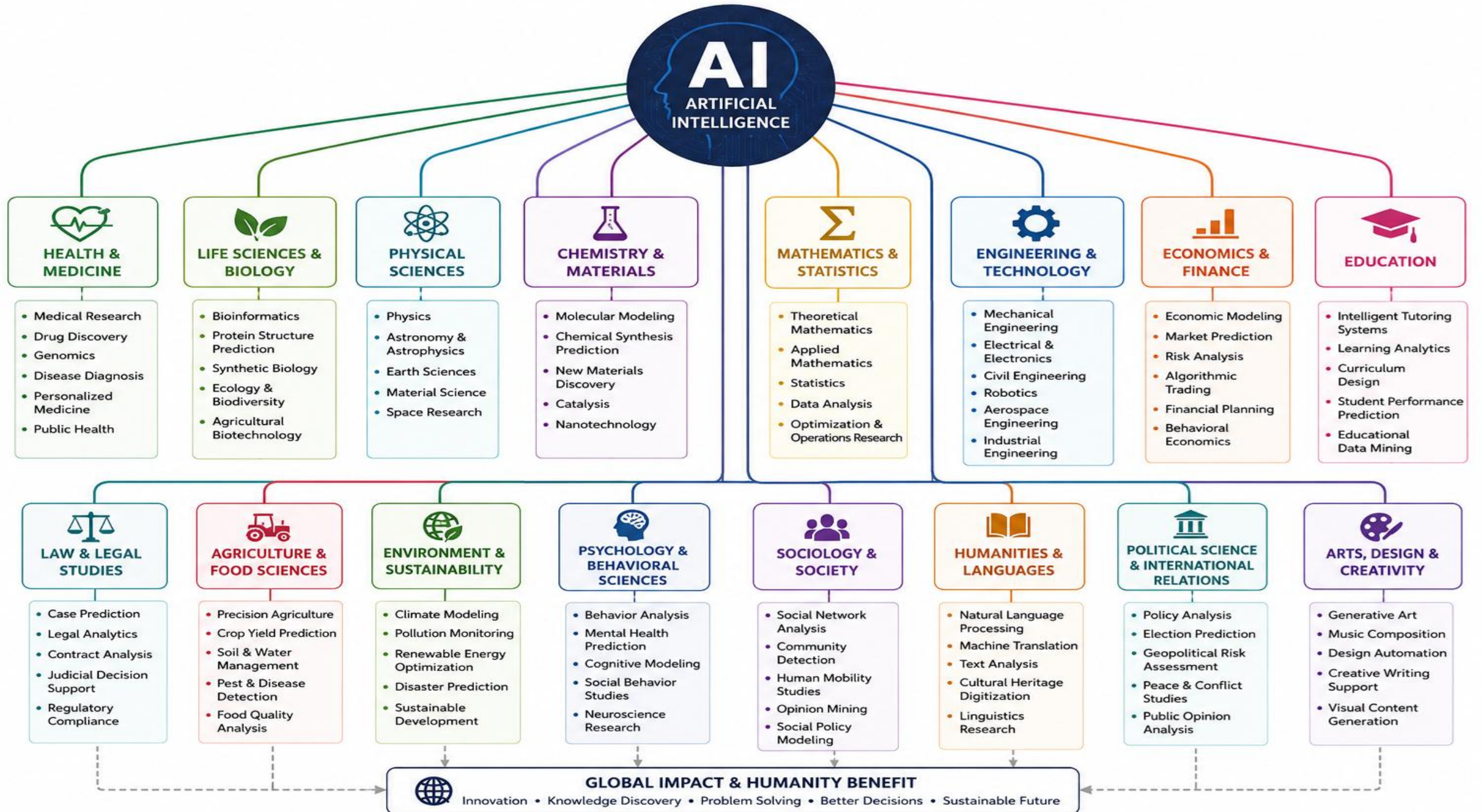
- The history of NLP generally starts in the 1950s, although work can be found from earlier periods. In 1950, Alan Turing published an article titled "Computing Machinery and Intelligence" which proposed what is now called the Turing test as a criterion of intelligence.
- During the 1970s many programmers began to write 'conceptual ontologies', which structured real-world information into computer-understandable data. Examples are MARGIE (Schank, 1975), SAM (Cullingford, 1978), PAM (Wilensky, 1978), TaleSpin (Meehan, 1976), QUALM (Lehnert, 1977), Politics (Carbonell, 1979), and Plot Units (Lehnert 1981). During this time, many chatterbots were written including PARRY, Racter, and Jabberwacky.
- Up to the 1980s, most NLP systems were based on complex sets of hand-written rules. Starting in the late 1980s, however, there was a revolution in NLP with the introduction of machine learning algorithms for language processing. This was due to both the steady increase in computational power (see Moore's Law) and the gradual lessening of the dominance of Chomskyan theories of linguistics (e.g. transformational grammar)

• Why NEP2020 Focus so much on Interdisciplinary and Language

- Language (engineering)
- Speech to text
- Speech recognition
- NLP
- A I
- Computer science
- Electronics
- Physics
- Math's
- Philosophy (Ontology - Epistemology)
- Inability to express knowledge in Language



AI AS ROOT FOR ALL RESEARCH AREAS



Artificial intelligence

- "The study and design of intelligent agents" in which an intelligent agent is a system that perceives its environment and takes actions that maximize its chances of success. John McCarthy, who has given this term in 1955, defines it as "the science and engineering of making intelligent machines".
- The central problems (or goals) of AI research include
 - Reasoning
 - Knowledge
 - Planning
 - Learning
 - Natural language processing
 - Perception
 - and the ability to move and manipulate objects

Other parameters and applications

- Logic
- Emotion
- Cognition
- Knowledge representation
- Pattern matching
- Psychology
- Philosophy
- Epistemology
- Ontology
- Economics
- Planning
- Rational
- Problem solving
- Unsupervised Learning

Logic : तर्क science uses in reasoning

- Propositional logic
- Predicates Logic – prolog ,lisp
- Fuzzy logics - Perception

Reasoning: decision making using logical thinking

- Logic + reasoning = rule development , fact production leading to knowledge base creation
- Monotonic reasoning
- Non monotonic reasoning (contradiction)

Knowledge Representation

- Building Knowledge Base
 - Semantic networks
 - Frames – procedural and declarative
 - Conceptual dependency
 - Scripts

Philosophy : knowledge : Truth :How : What

- Epistemology :How
 - Study of nature
- Ontology : metaphysics : Information science :What
 - Essence of things
 - Stemming
 - Synonyms

the nature of existence? When we ask deep questions about "what is the nature of the universe?" or "Is there a god?" or "What happens to us when we die?" or "What principles govern the properties of matter?" we are asking inherently ontological questions.

Epistemology: The branch of philosophy concerned with the nature of knowledge itself, its possibility, scope, and general basis. More broadly: *How do we go about knowing things? or How do we separate true ideas from false ideas? or How do we know what is true? or "How can we be confident when we have located 'truth'?" "What are the systematic ways we can determine when something is good or bad?"*

So ontology is about *what* is true and epistemology then is about *methods* of figuring out those truths.

The split between Plato and Aristotle is both ontological and epistemic. The split between religion and science is both ontological and epistemic. For example, religion and science offer two very different ontologies (theories about what is out there) and epistemology (ways to figure out what is out there). And the split between Plato and Aristotle matches exactly the split between religion and science...and you should leave this class understanding why and how! See [Plato vs. Aristotle](#)

- Ontology is the philosophical field revolving around (the study of) the nature of reality (all that is or exists), and the different entities and categories within reality.
- Epistemology is the philosophical field revolving around (the study of) knowledge and how to reach it. One might say that it includes the ontology of knowledge.
- Examples of theories within the field of ontology are: ontological monism, pluralism, idealism, materialism, dualism, etc.
- Examples of theories within the field of epistemology are: realism, relativism, rationalism, irrationalism, etc.

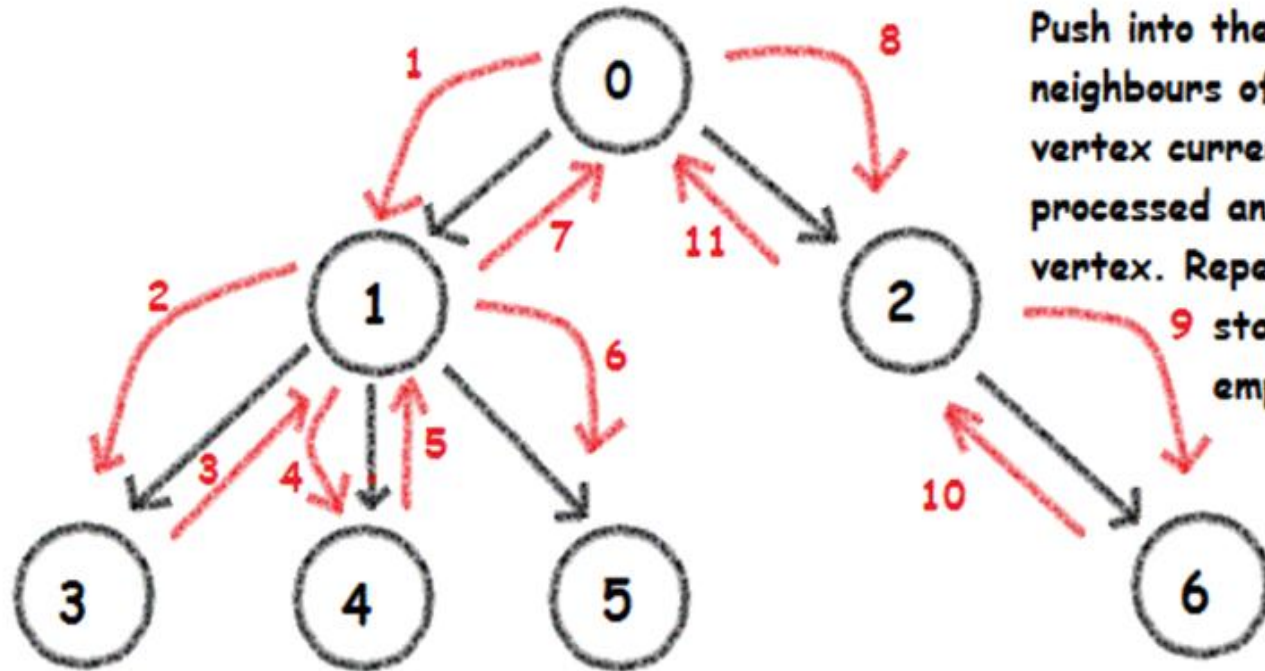
Cognition : Knowledge processing through sense organs , environment , experiences .

- Can we implement cognition in machines ?

Heuristics

- AI problem solving algorithms
 - blind search
 - Brute force search
 - DFS
 - BFS
 -
 -
 -
 -
 -
- Heuristic search
 - Generate and test
 - Hill climbing
 - feedback
 - A star
 - Best first search
 - Greedy Algorithm
 - And OR search

Red arrows indicate the order of search.



Push into the stack the neighbours of the vertex currently being processed and Pop the vertex. Repeat until stack is not empty.

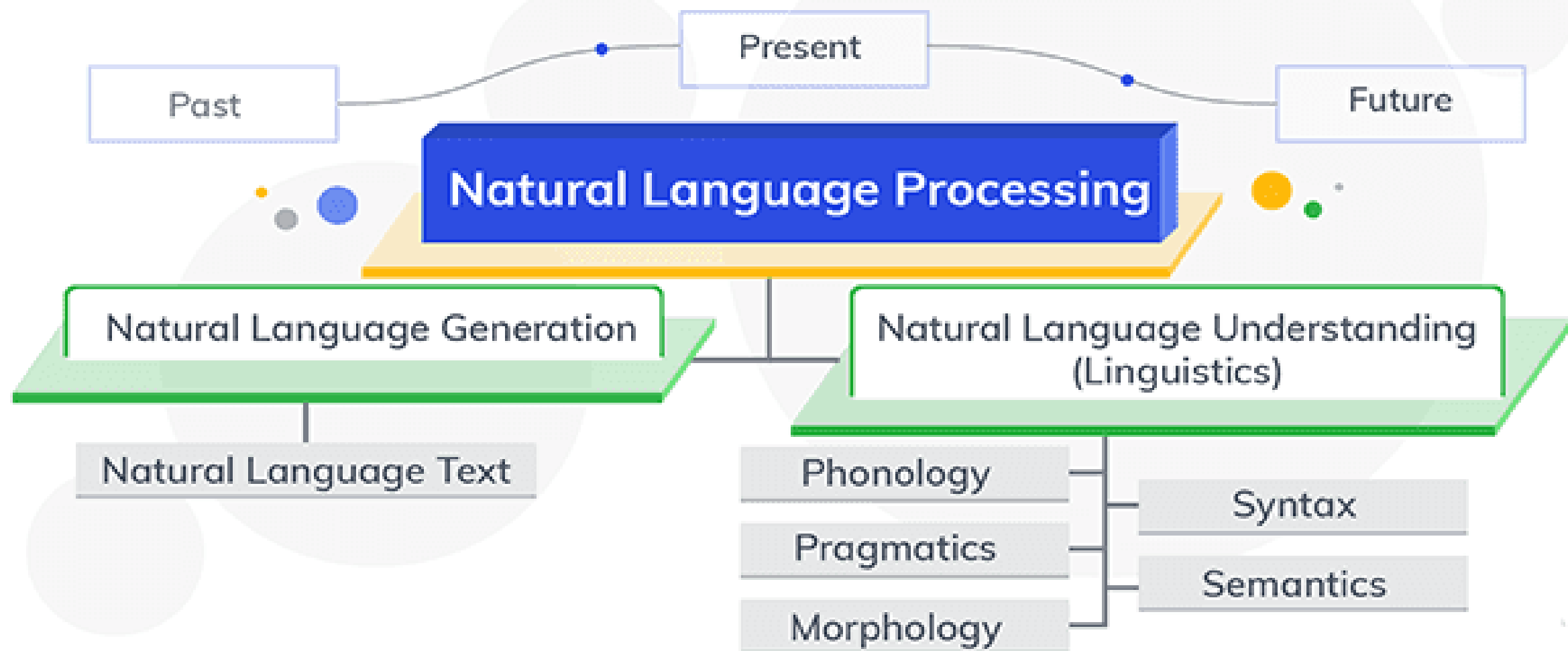
Vertex	Stack
	0
0	1, 2
1	3, 4, 5, 2
3	4, 5, 2
4	5, 2
5	2
2	6
6	

Depth First Search

Natural language processing

- NLP technically known as CL is applied area of AI and CS which is used for natural language manipulation by computers in the same way as done by humans.
- Natural language generation
How language generated and can a machine generate language using big data , information?
- Natural language understanding
- Natural language learning

Evolution of NLP



Phonology – This science helps to deal with patterns present in the sound and speeches related to the sound as a physical entity. Word base consonants vowels

Lexical

Pragmatics – Analysis involve the study of language in its contexts of use.(context based meaning)

Morphology – This science deals with the structure of the words and the systematic relations between them internal structure of word minimum meaningful meaning

Syntax – This science deal with the structure of the sentences.

Semantics – This science deals with the literal meaning of the words, phrases as well as sentences.

Pragmatics and Discourse Analysis involve the study of language in its contexts of use. **Pragmatics** focuses on the effects of context on meaning, **and Discourse** Analysis studies written **and** spoken language in relation to its social context.



Lexical Analysis

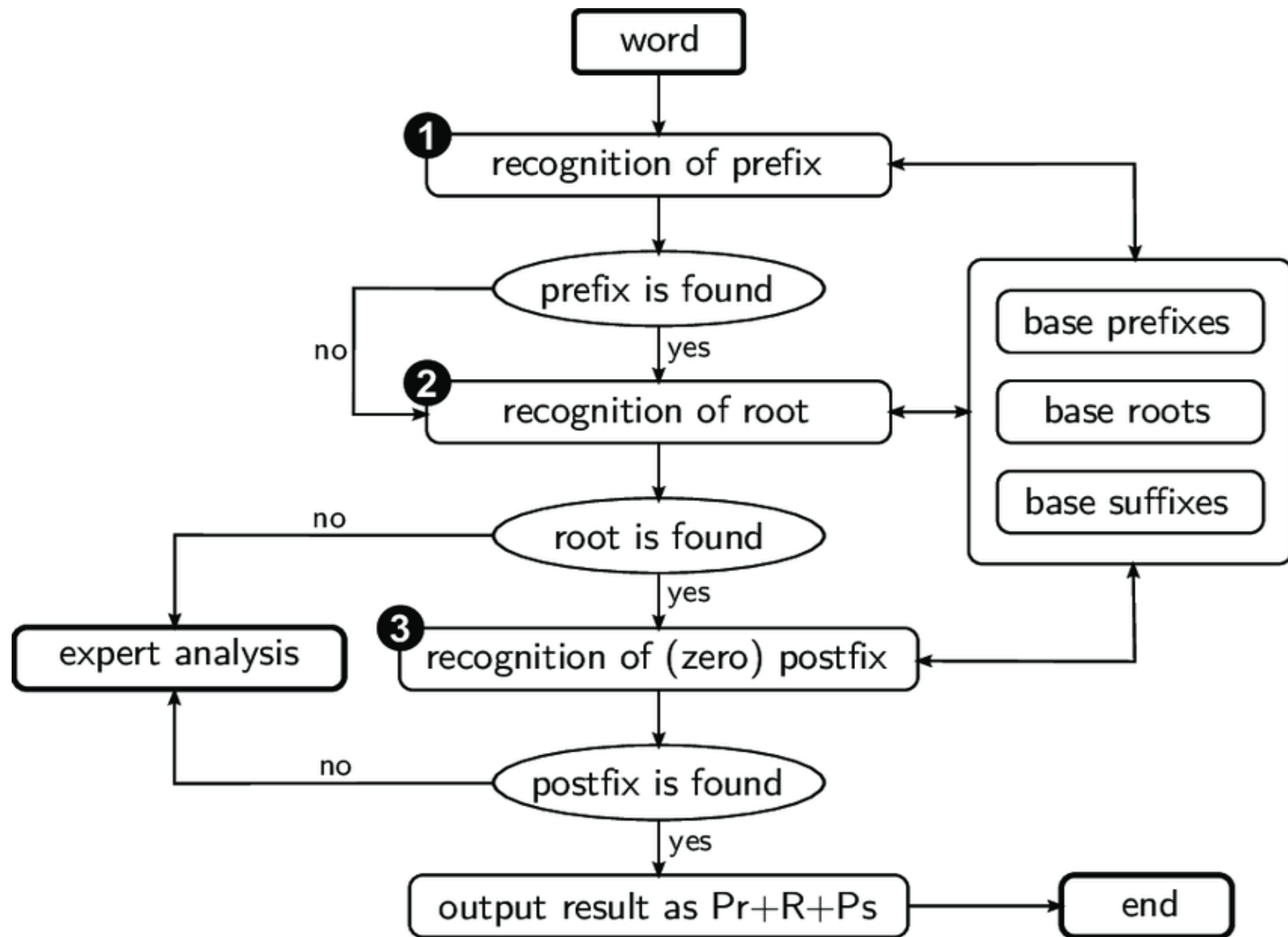
- Recognize tokens and ignore white spaces, comments

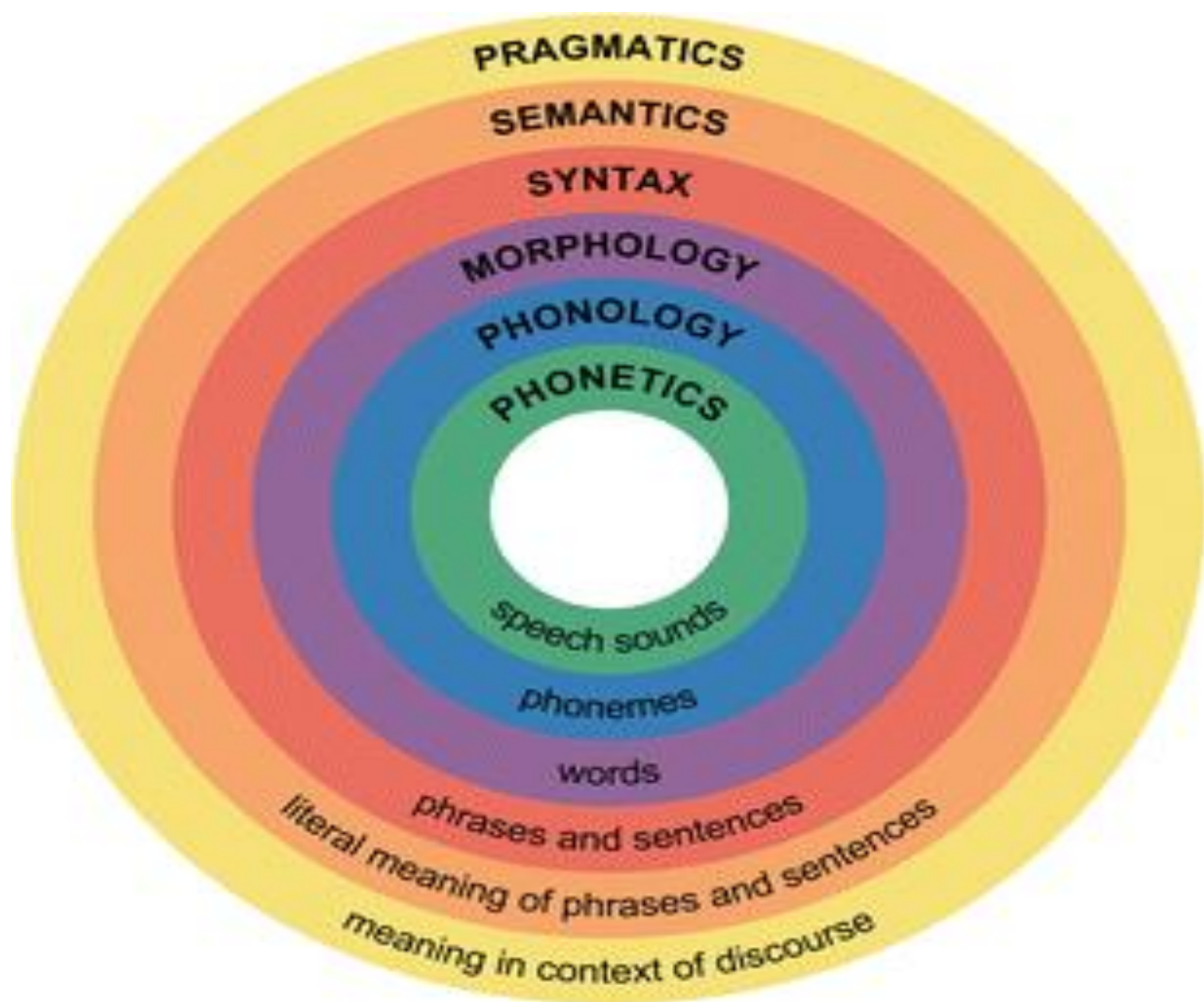
i	f		(	x	1		*	x	2	<	1	.	0	)	{
---	---	--	---	---	---	--	---	---	---	---	---	---	---	---	---

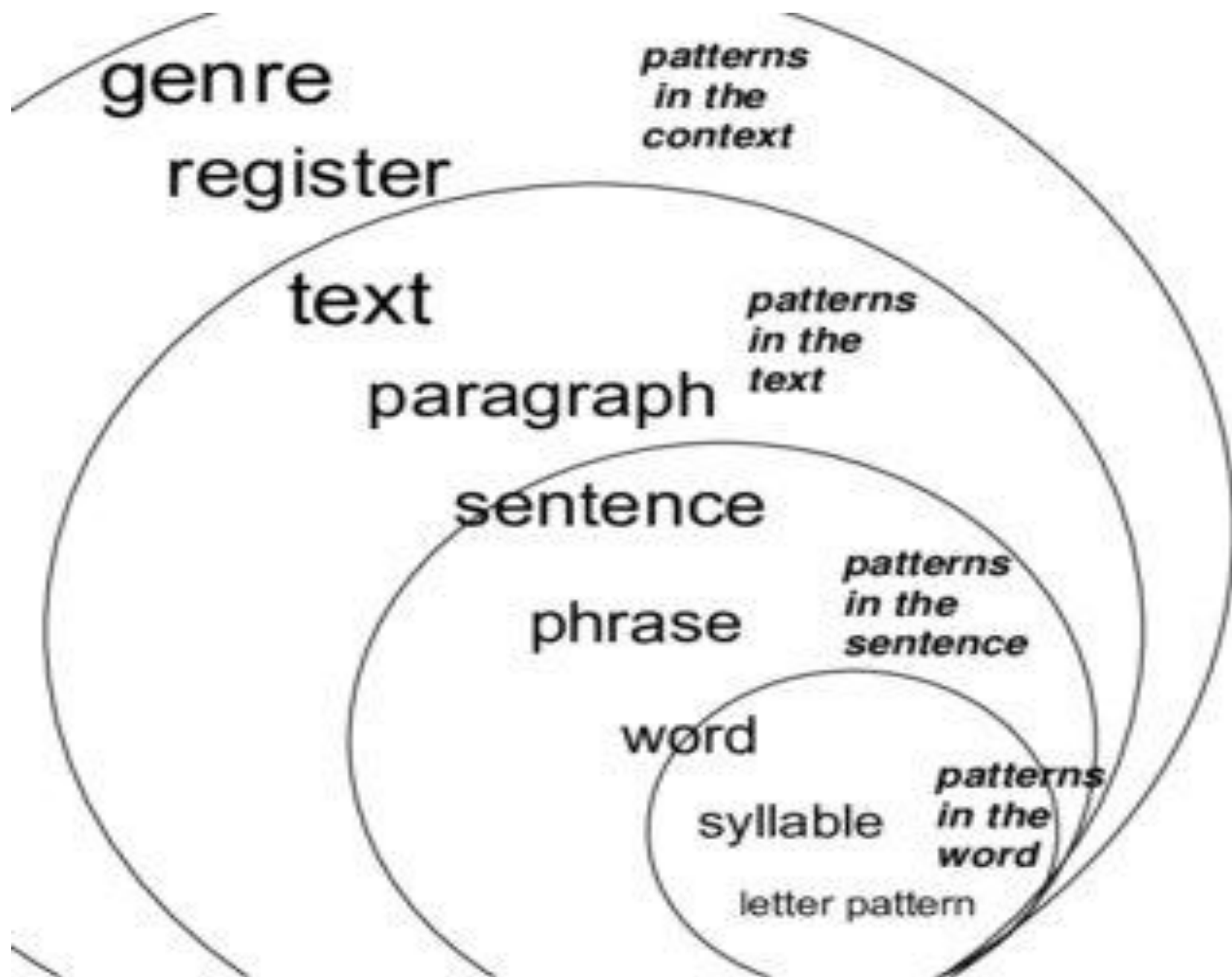
Generates token stream

if	(	x1	*	x2	<	1.0	)	{
----	---	----	---	----	---	-----	---	---

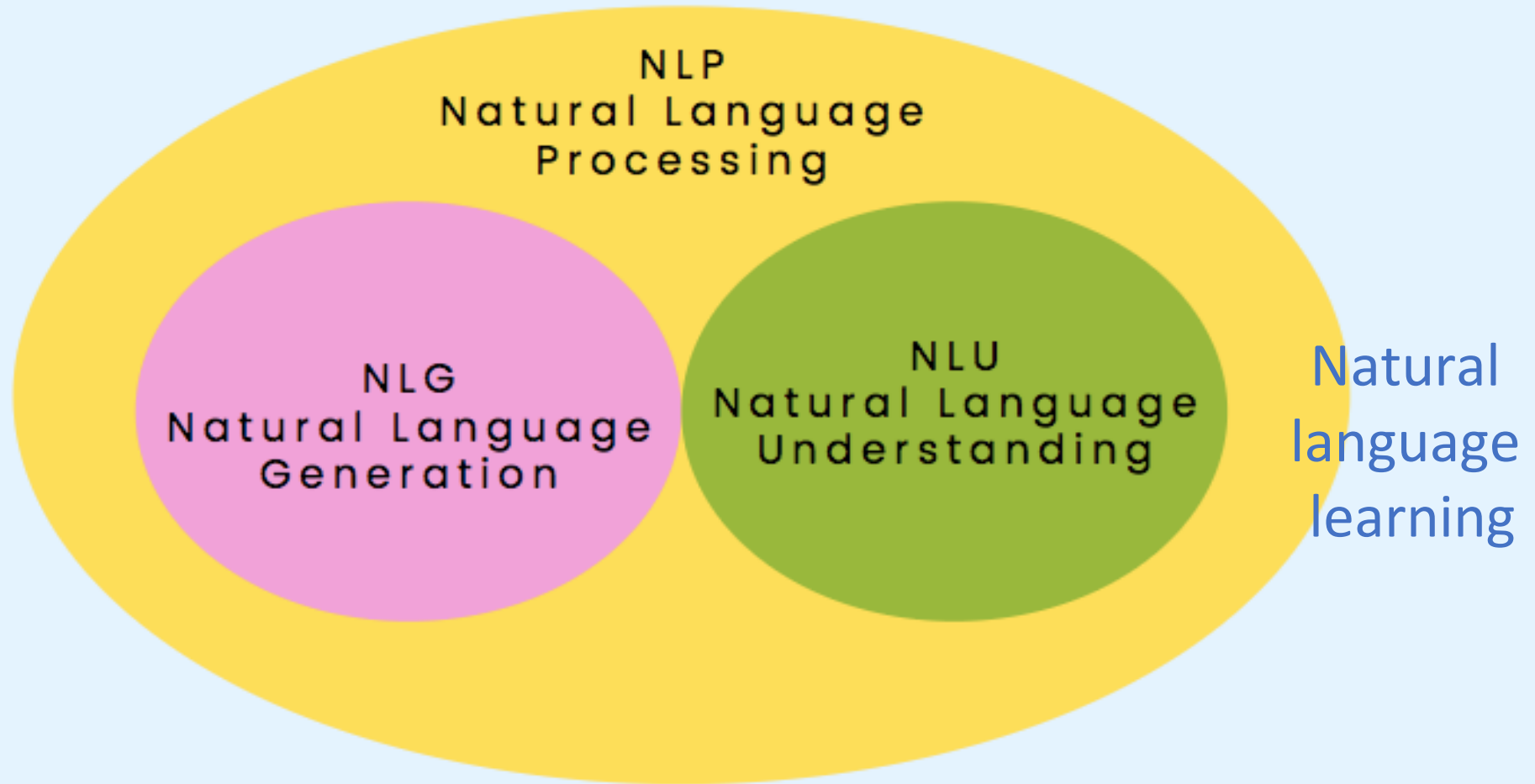
- Error reporting
- Model using regular expressions
- Recognize using Finite State Automata



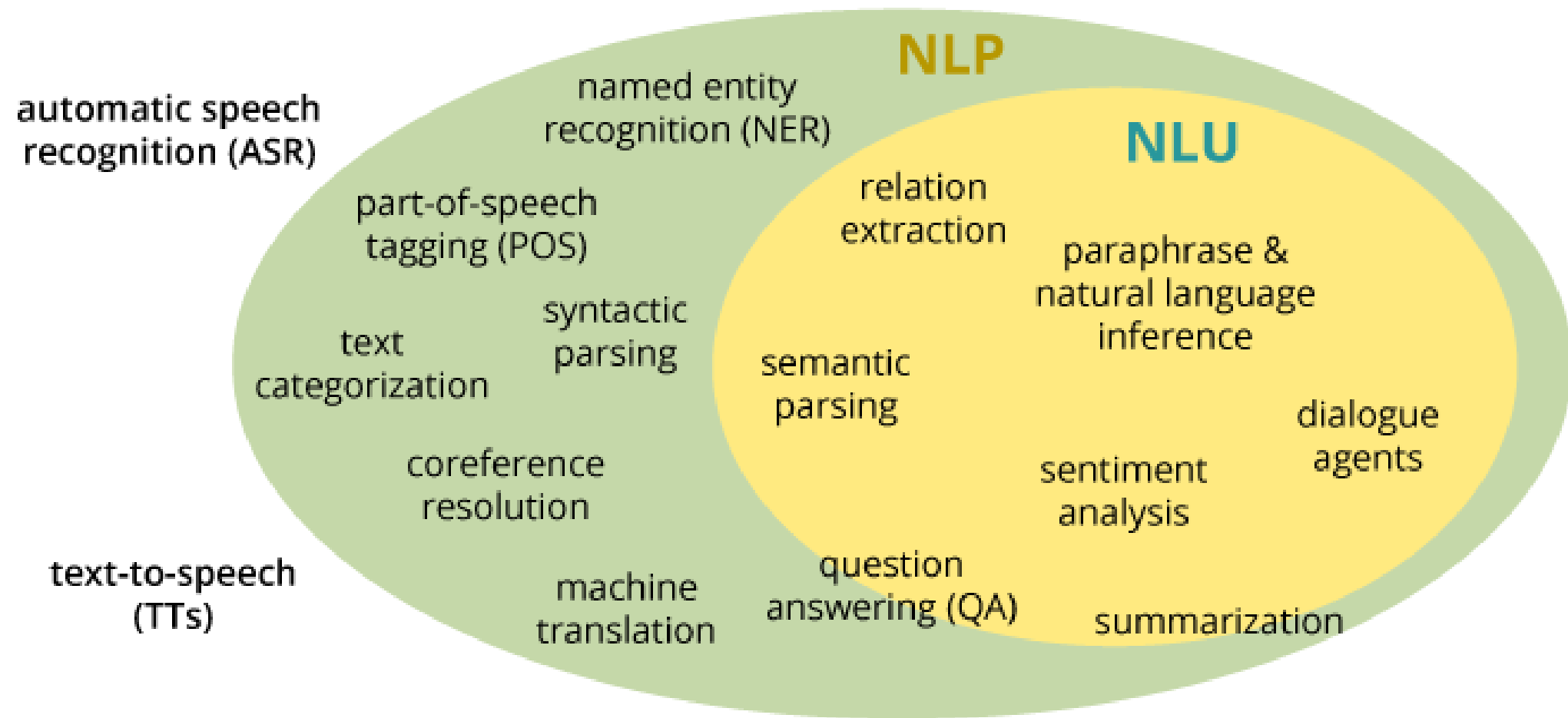


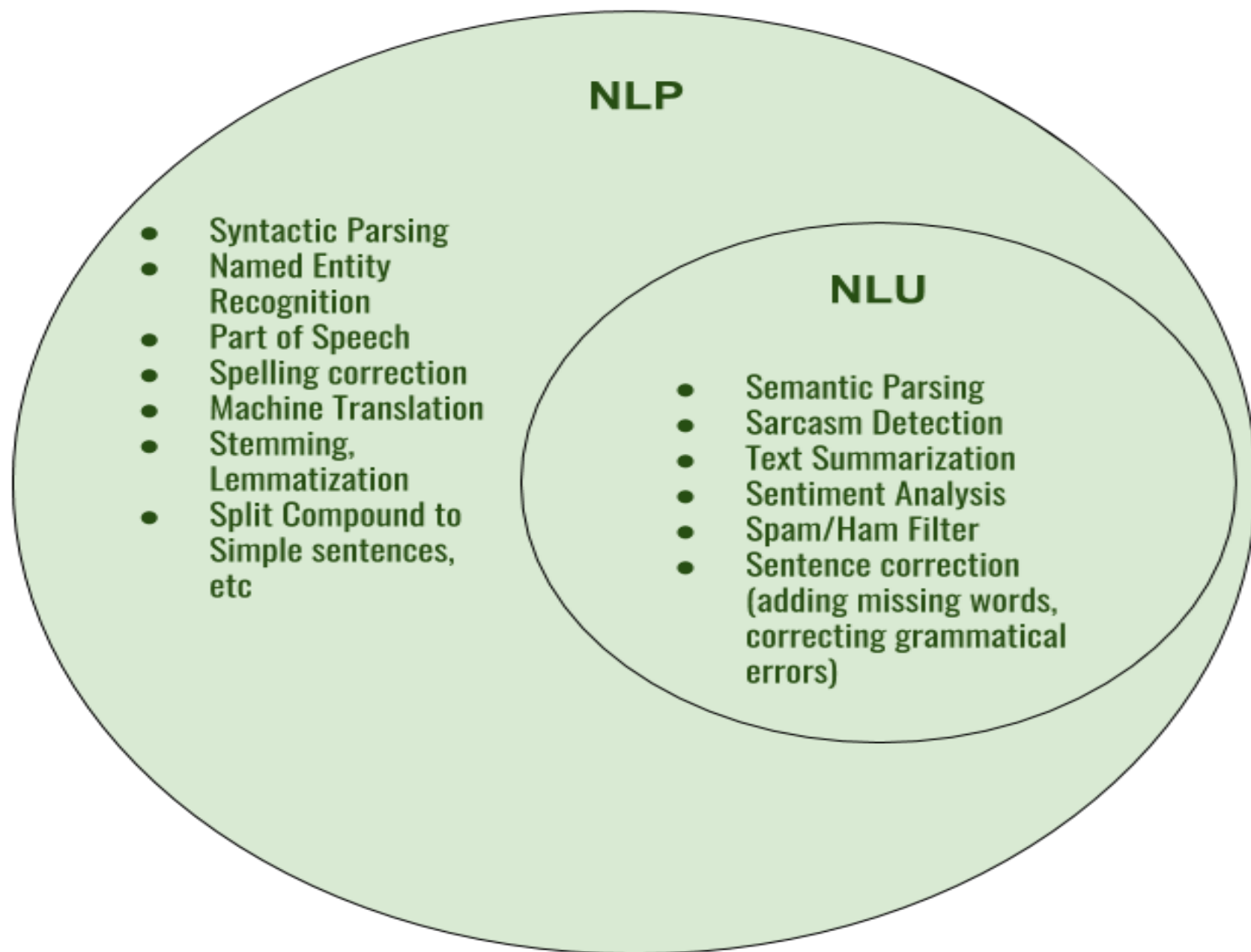


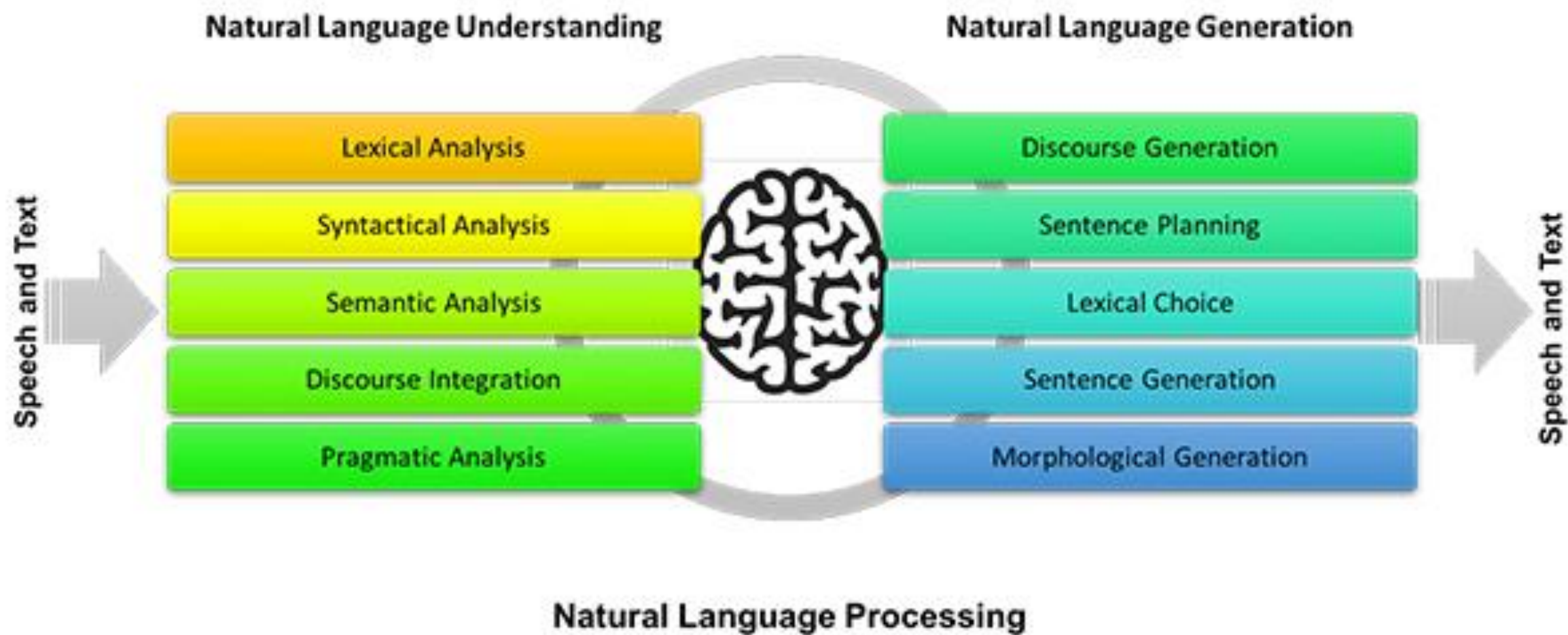
Natural Language Processing



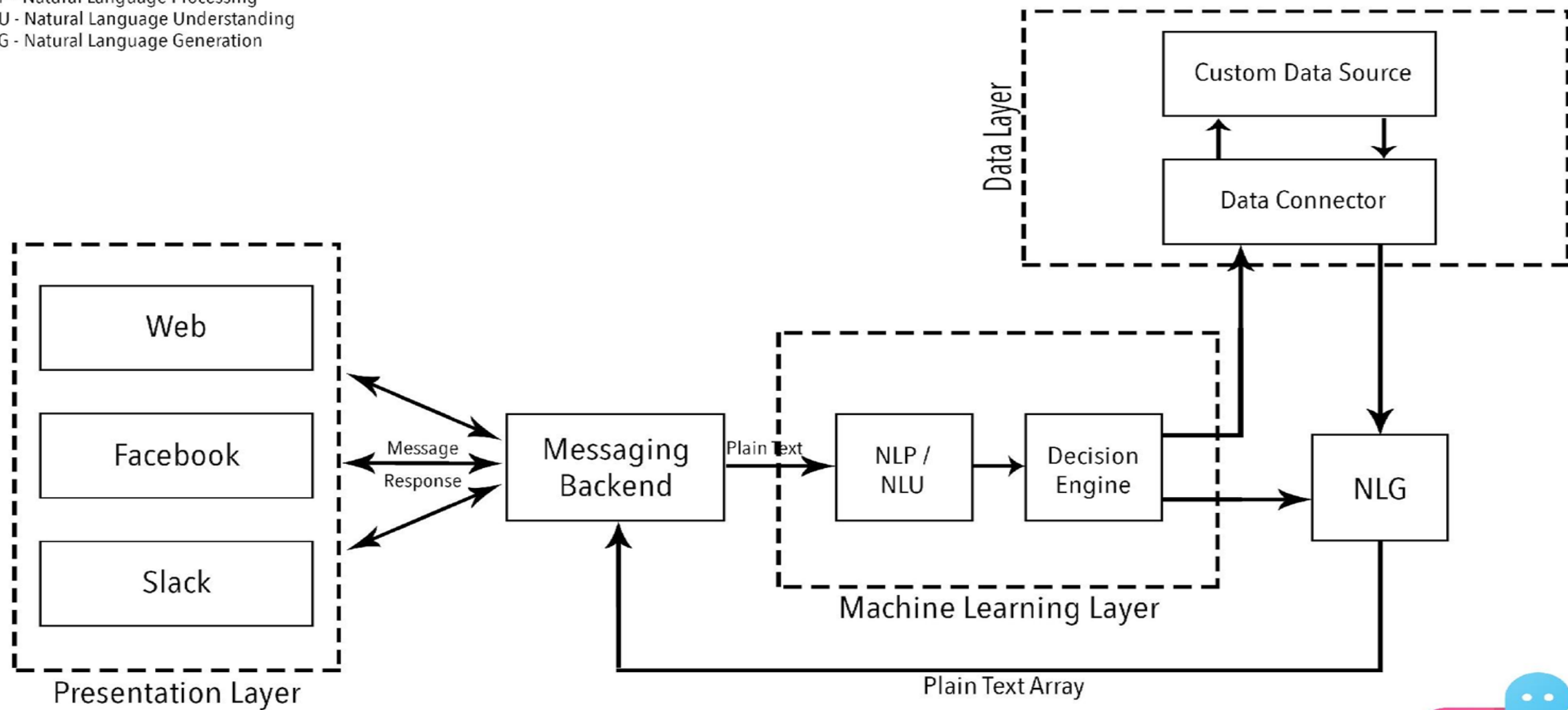
Terminology: NLU vs. NLP vs. ASR







Key:
NLP - Natural Language Processing
NLU - Natural Language Understanding
NLG - Natural Language Generation



Pre - processing in NLP

- Splitting
- Chunking
- Structurization
- Organizing

Segmentation

- Speech segmentation,
- Text segmentation,
- Topic segmentation,
- Word segmentation

Tokenization

- Words as tokens
- Sentences as tokens
- POS tokens
- Bag of words
- Domains as tokens ETC
- Others



Stop words

- from nltk.corpus import stopwords
sw = stopwords.words("english")
- ["i", "me", "my", "myself", "we", "our", "ours", "ourselves", "you", "your", "yours", "yourself", "yourselves", "he", "him", "his", "himself", "she", "her", "hers", "herself", "it", "its", "itself", "they", "them", "their", "theirs", "themselves", "what", "which", "who", "whom", "this", "that", "these", "those", "am", "is", "are", "was", "were", "be", "been", "being", "have", "has", "had", "having", "do", "does", "did", "doing", "a", "an", "the", "and", "but", "if", "or", "because", "as", "until", "while", "of", "at", "by", "for", "with", "about", "against", "between", "into", "through", "during", "before", "after", "above", "below", "to", "from", "up", "down", "in", "out", "on", "off", "over", "under", "again", "further", "then", "once", "here", "there", "when", "where", "why", "how", "all", "any", "both", "each", "few", "more", "most", "other", "some", "such", "no", "nor", "not", "only", "own", "same", "so", "than", "too", "very", "s", "t", "can", "will", "just", "don", "should", "now"]

NLP TOOLKITS

- BORIS
- CMU Sphinx
- Corpora
- GATE-8.1
- Speaktait
- LOLITA
- Snow ball
- Festival speech synthesis system
- CTAKES
- Maluuba
- Regulus Grammar Compiler
- Never ending language learning
- Scala NLP
- JGibLDA- v1.0
- ETAP-3
- Apache lucene core
- Stanford NLP
- Chat GPT
- Spacy
- Text blob

NLP TOOLKITS w.r.t. Programming Language

- DELPH-IN (CH,LISP)
- Deeplearning4 (Scala and also Java)
- Mallet (Java)
- Natural language toolkit(python)
- Lingua stream(Java)
- Distinguo(C++)
- Modular Audio Recognition Framework(Java)
- Monty Lingua(Python and also Java)
- Gensim (python based)
- TextBlob(Python)

Four Fundamental Approaches of NLP

- 1. Vocabulary / Dictionary / Mix Matching /Lexical
- 2. Mathematical /Statistical / Probabilistic / ML /DL
- 3. Rule Base
- 4. Hybrid

Stemming

	original_word	stemmed_words
0	connect	connect
1	connected	connect
2	connection	connect
3	connections	connect
4	connects	connect

	original_word	stemmed_word
0	trouble	troubl
1	troubled	troubl
2	troubles	troubl
3	troublesome	troublesom

Ing
Ed
Ive
Ble
Ous
Ly
Ions
Ize
Ate
Ion
S
e

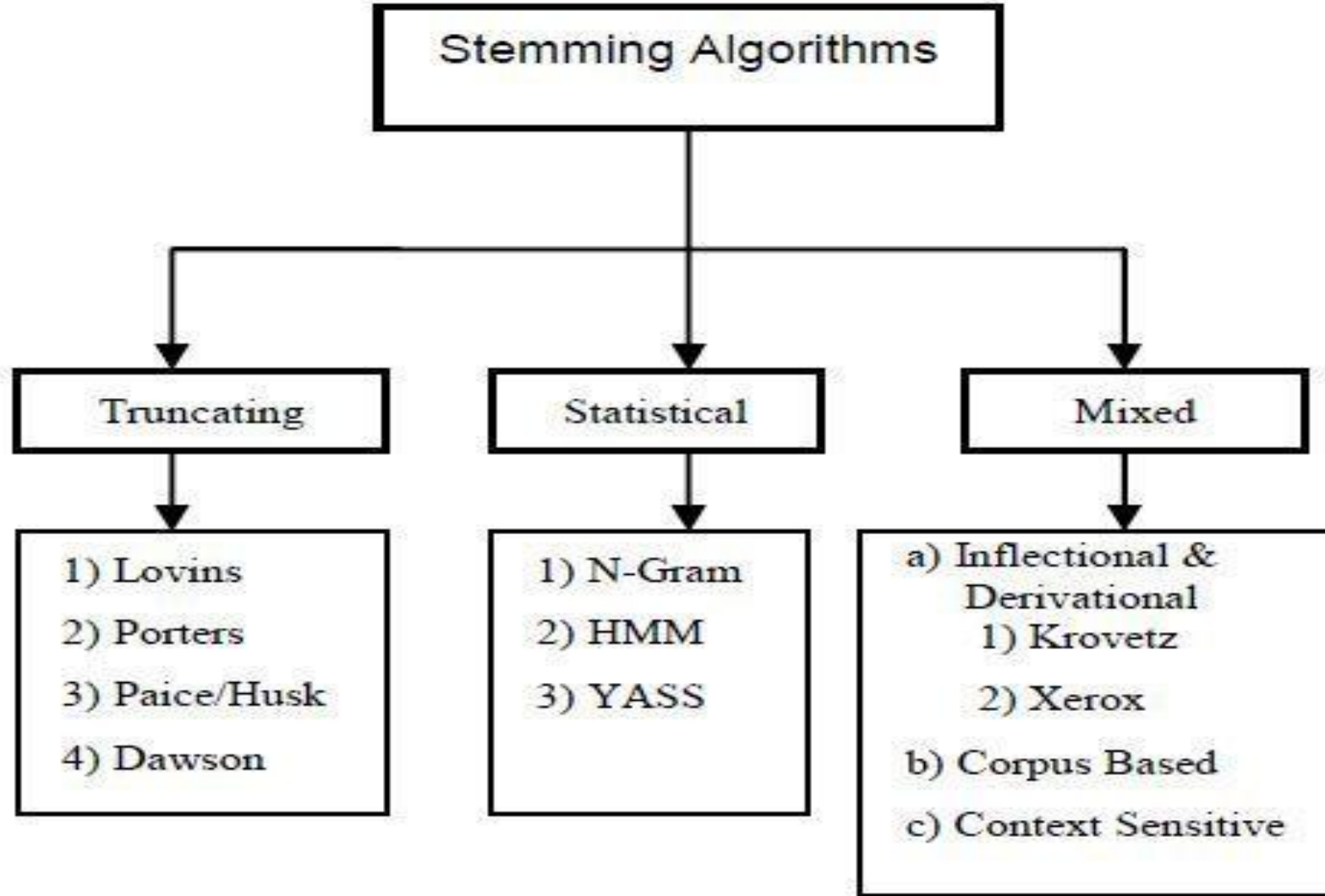
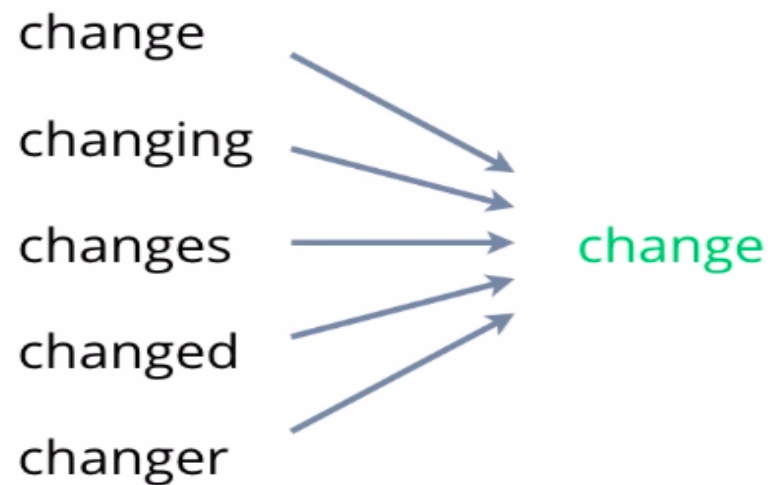
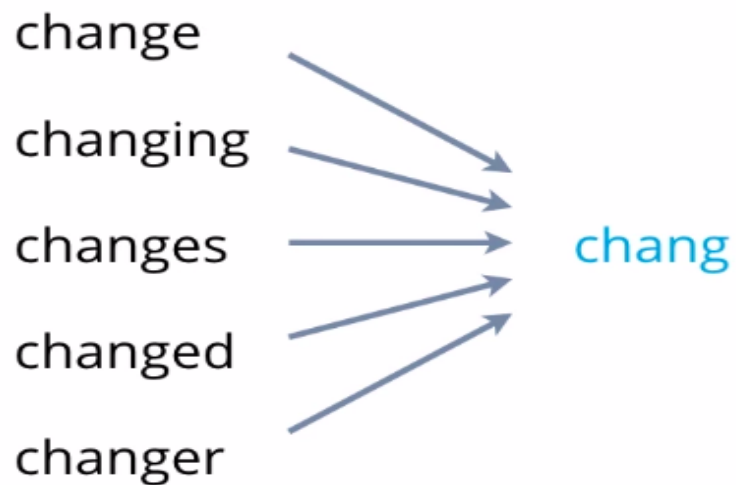


Figure 1. Types of stemming algorithms

Lemmatization (lemma)

*WORDNET *NLTK

Stemming vs Lemmatization



Both works on root word Algo. But lemmatization

- Can change better best to GOOD as it is context driven

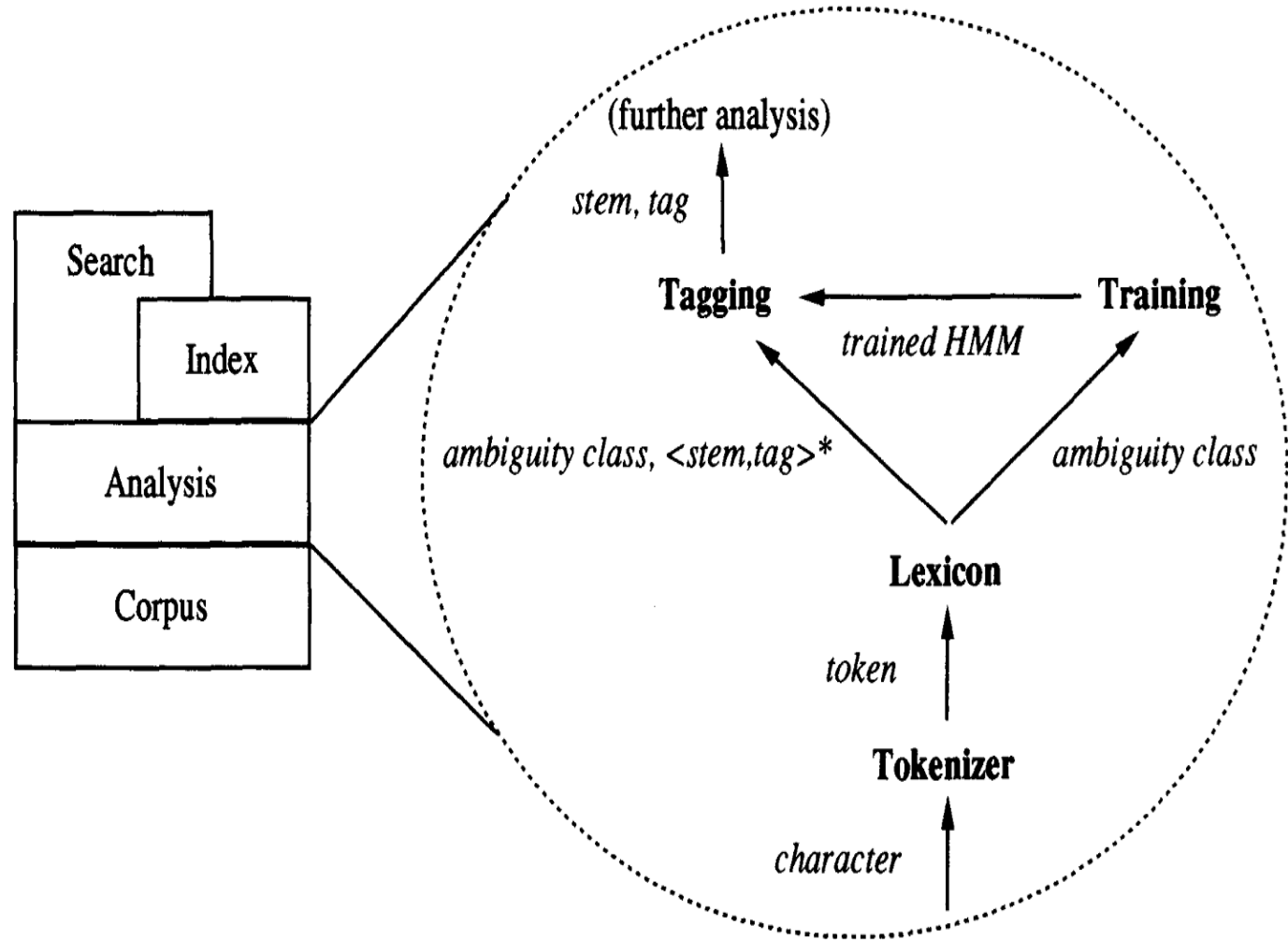
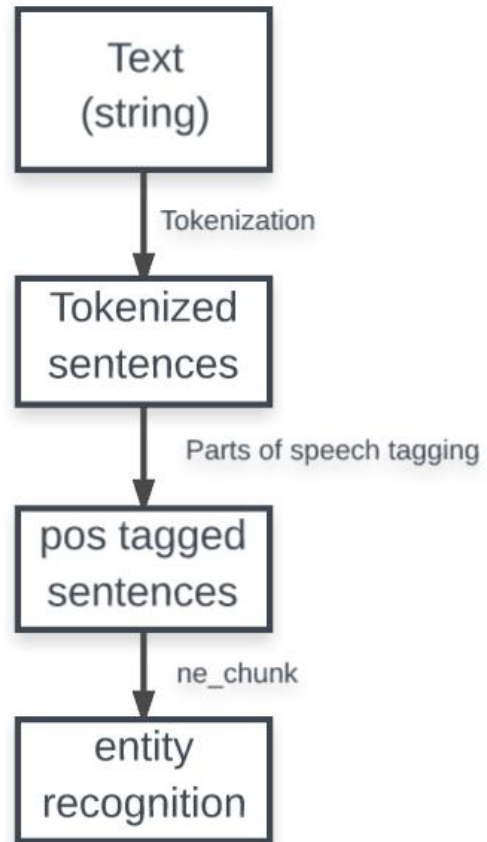
Stemming

adjustable → adjust
formality → formaliti
formaliti → formal
airliner → airlin △

Lemmatization

was → (to) be
better → good
meeting → meeting

POS Taggers



Part of Speech (POS)

Noun fish, book, house, pen, procrastination, language

Proper noun John, France, Barack, Goldsmiths, Python

Verb loves, hates, studies, sleeps, thinks, is, has

Adjective grumpy, sleepy, happy, bashful

Adverb slowly, quickly, now, here, there

Pronoun I, you, he, she, we, us, it, they

Preposition in, on, at, by, around, with, without

Conjunction and, but, or, unless

Determiner the, a, an, some, many, few, 100

Part of speech tags¹

CC - Coordinating conjunction
CD - Cardinal number
DT - Determiner
EX - Existential there
FW - Foreign word
IN - Preposition or subordinating conjunction
JJ - Adjective
JJR - Adjective, comparative
JJS - Adjective, superlative
NN - Noun, singular or mass
NNS - Noun, plural
NNP - Proper noun, singular
NNPS - Proper noun, plural
PDT - Predeterminer
NP - Noun Phrase.
PP - Prepositional Phrase
VP - Verb Phrase.

PRP - Personal pronoun
RB - Adverb
RBR - Adverb, comparative
RBS - Adverb, superlative
RP - Particle
SYM - Symbol
TO - to
UH - Interjection
VB - Verb, base form
VBD - Verb, past tense
VBG - Verb, gerund or present participle
VBN - Verb, past participle
VBP - Verb, non-3rd person singular present
VBZ - Verb, 3rd person singular present
WDT - Wh-determiner
WP - Wh-pronoun
WRB - Wh-adverb

¹ <http://bulba.sdsu.edu/jeanette/thesis/PennTags.html>

be further classified into the following types.

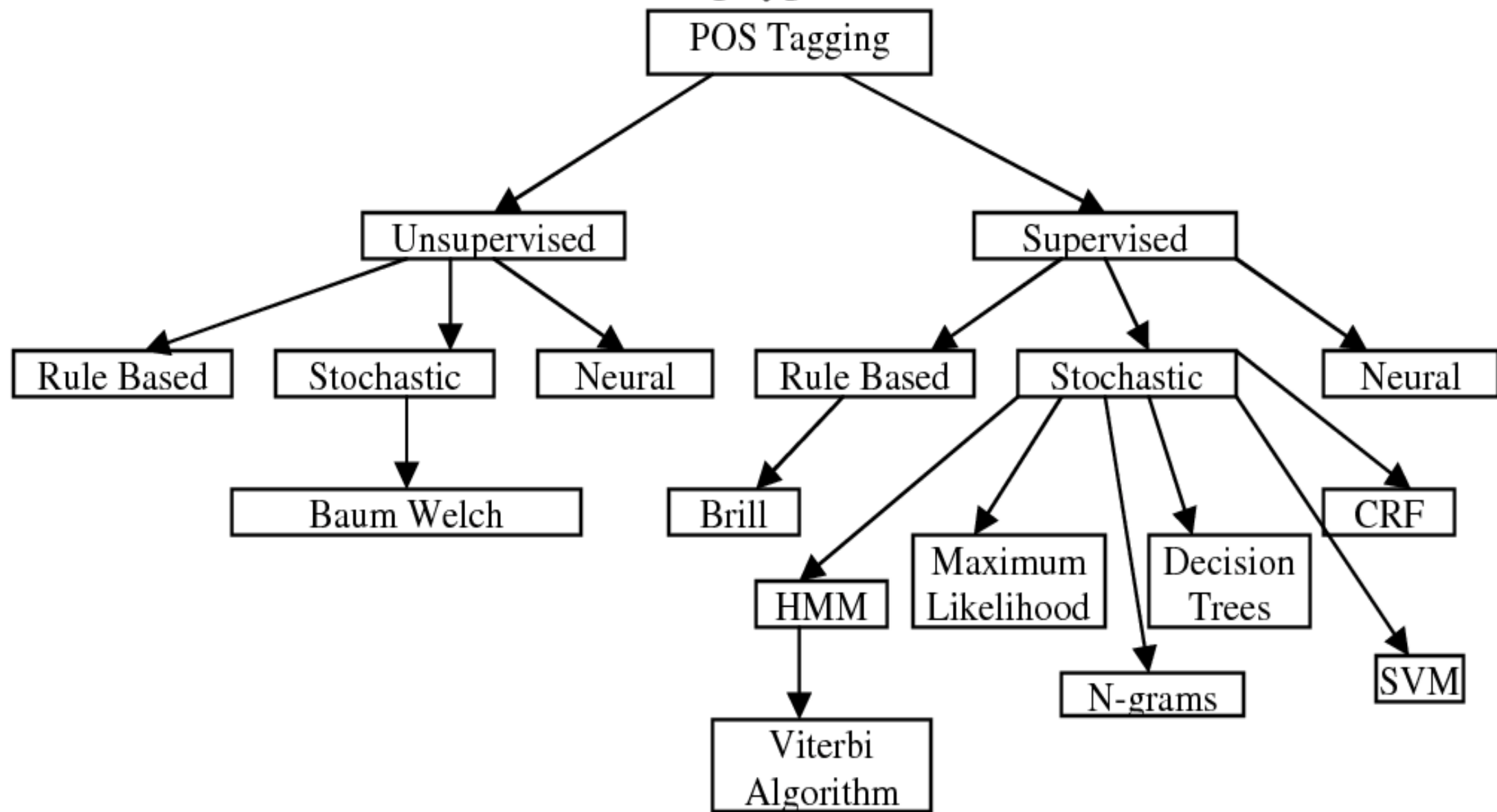


Figure 1 Various techniques for POS tagging

Rule #1

A noun is a person, place, thing or idea. A verb is an action word.
The bird flew.

Rule #2

A complete sentence must include a subject and a predicate.
The angry bird flew quickly across the sky.

Rule #3

The only exception to the above rules is the imperative sentence.
Go away!

Rule #4

Adjectives can go directly before the noun they describe, or after it, if separated by a verb.
• The angry bird flew.
• The bird is angry.

Rule #5

A compound subject includes two or more simple subjects.
The bird and the plane flew.

Rule #6

A compound predicate includes two or more predicates.
The bird flew and sang.

Rule #7

A compound sentence includes more than one subject or predicate.
The bird sang and the plane flew.

Rule #8

An independent clause consists of a subject and a predicate, like a complete sentence.
The plane flew.

Rule #9

A dependent clause cannot form a complete sentence without additional words.
The plane flew when the bird sang.

Rule #10

The direct object is the noun being acted on by the verb.
The bird ate seeds.

Rule #11

The indirect object is the noun which receives the direct object.
The bird gave the seeds to me.

Rule #12

When written in passive voice, the object of the verb becomes the subject of the sentence.
The bird ate seeds.
--> The seeds were eaten by the bird.

Rule #13

Conjugations of the verb "to be" are essential to passive voice.
The seeds are being eaten by the bird.

Rule #14

Sentences written in passive voice can omit the subject of the acting verb.
The seeds were eaten.

Rule #15

In passive voice, the subject of the acting verb is connected by a prepositional phrase.
The bird ate seeds after lunchtime.

Rule #16

Prepositional phrases are sometimes separated by commas.
The bird, which belonged to my mother, ate seeds.

Rule #17

Adverbs function like prepositions.
The bird ate seeds quickly.

Rule #17

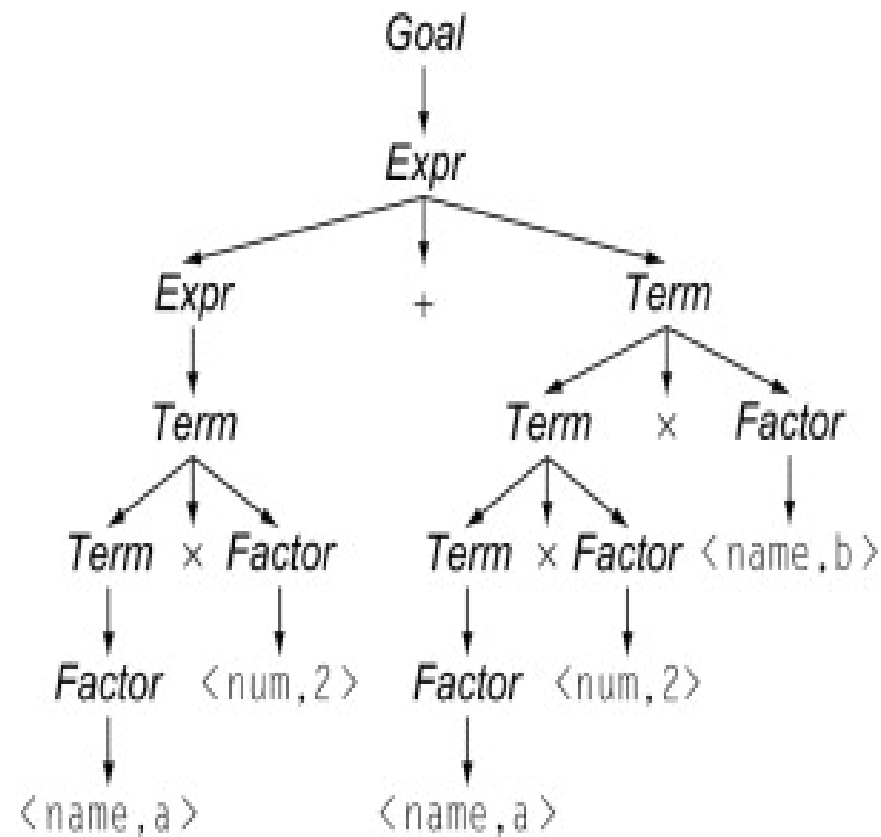
A subject compliment describes the subject of the sentence.
That bird is angry.



Parsing

- Compiler design VS NLP (syntax , operators vs NP VP)
- Deep Parsing: Shallow Parsing (meaning, relationship, POS)
- Chunking
- Syntax tree
- Grammatical rules
- Semantic tree ?

$Goal \rightarrow Expr$
 $Expr \rightarrow Expr + Term$
 $\quad | Expr - Term$
 $\quad | Term$
 $Term \rightarrow Term \times Factor$
 $\quad | Term \div Factor$
 $\quad | Factor$
 $Factor \rightarrow (Expr)$
 $\quad | num$
 $\quad | name$

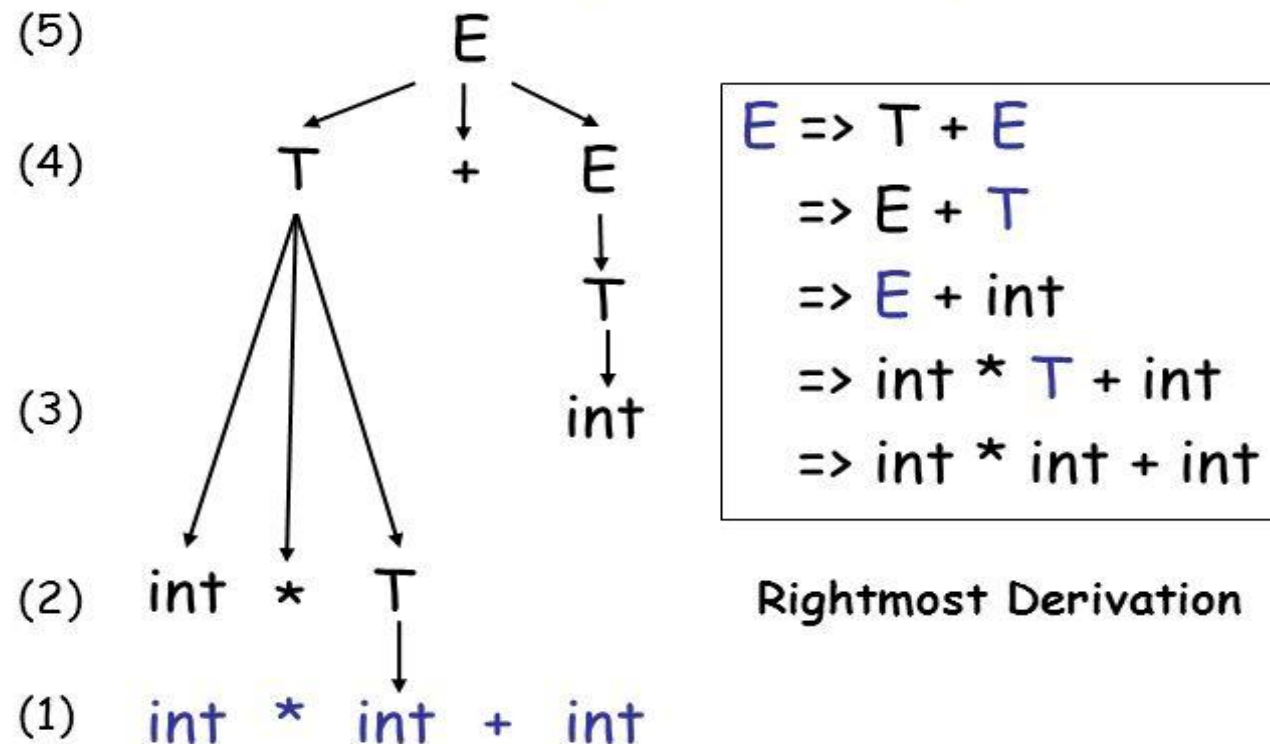


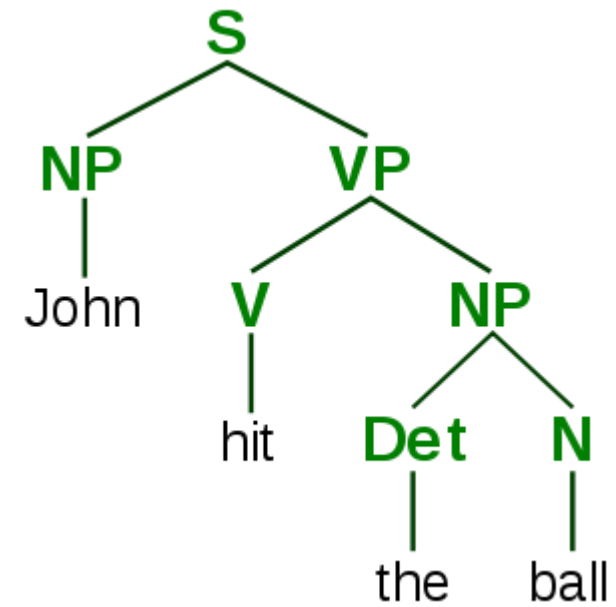
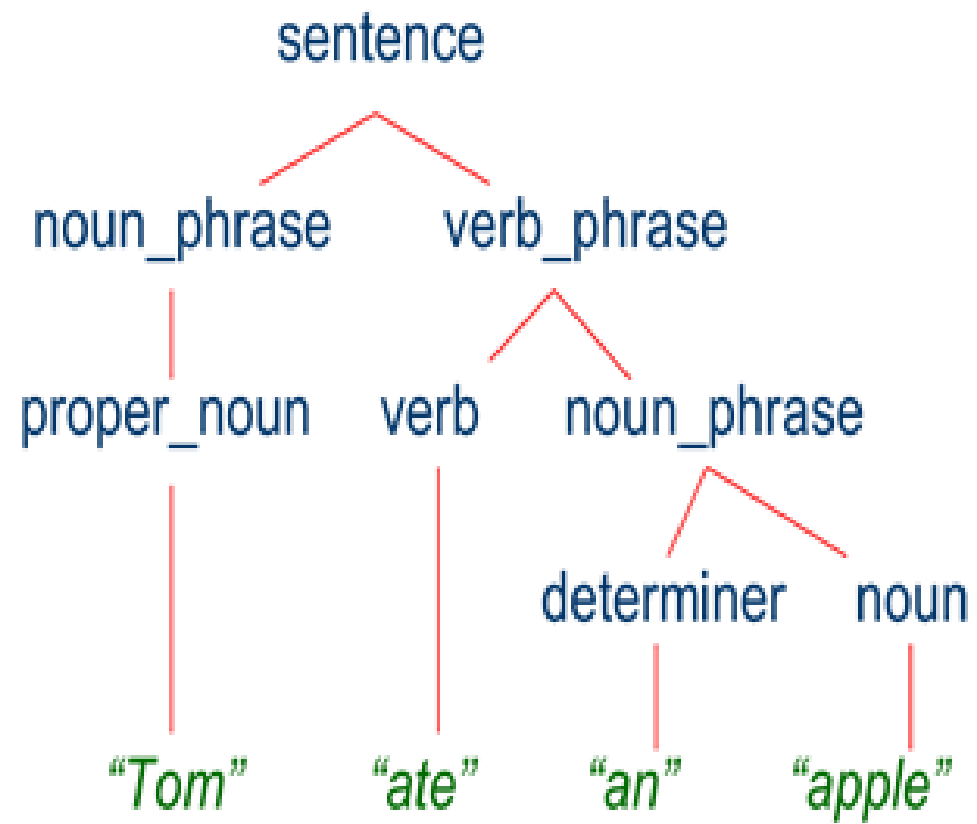
(a) Classic Expression Grammar

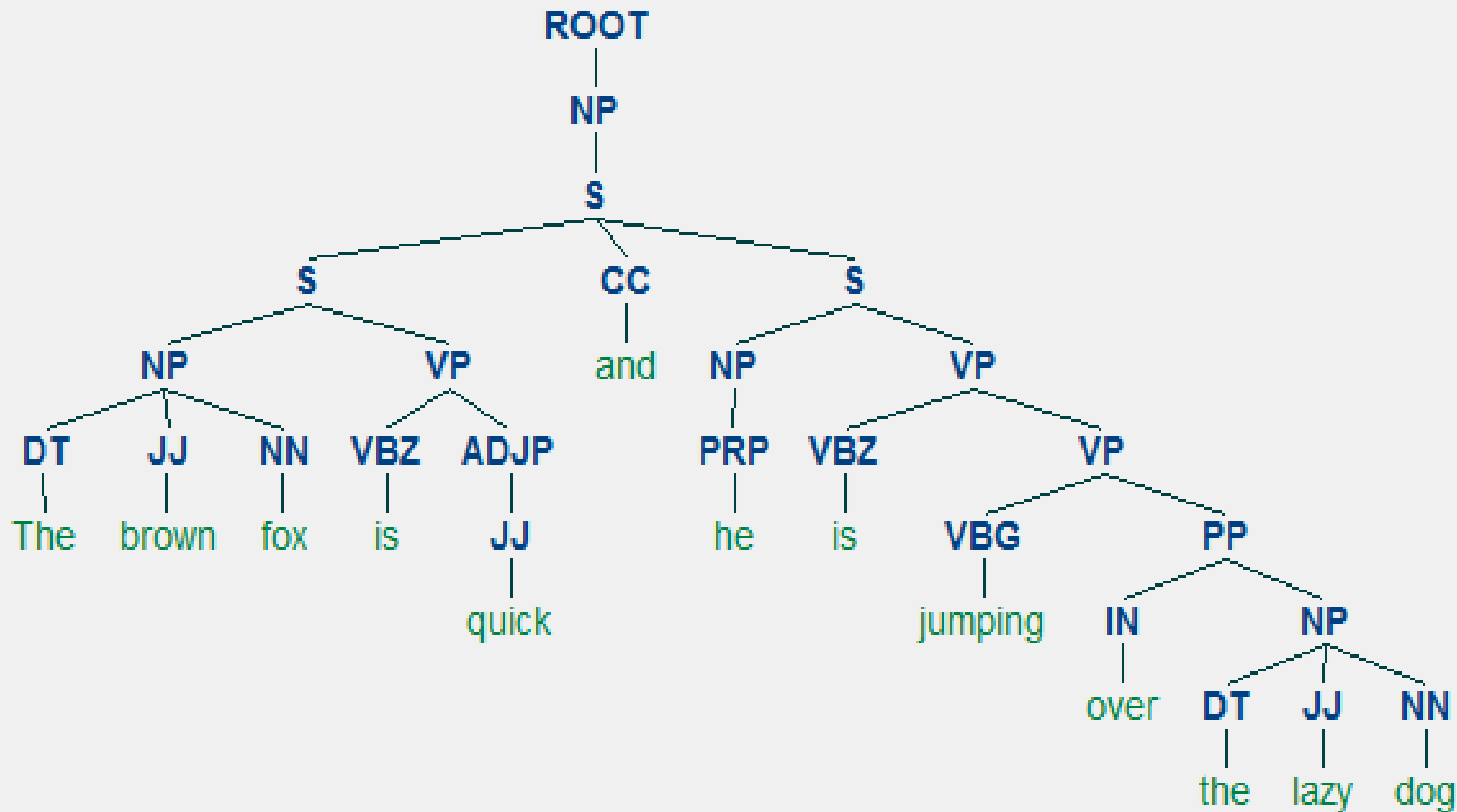
(b) Parse Tree for $a \times 2 + a \times 2 \times b$

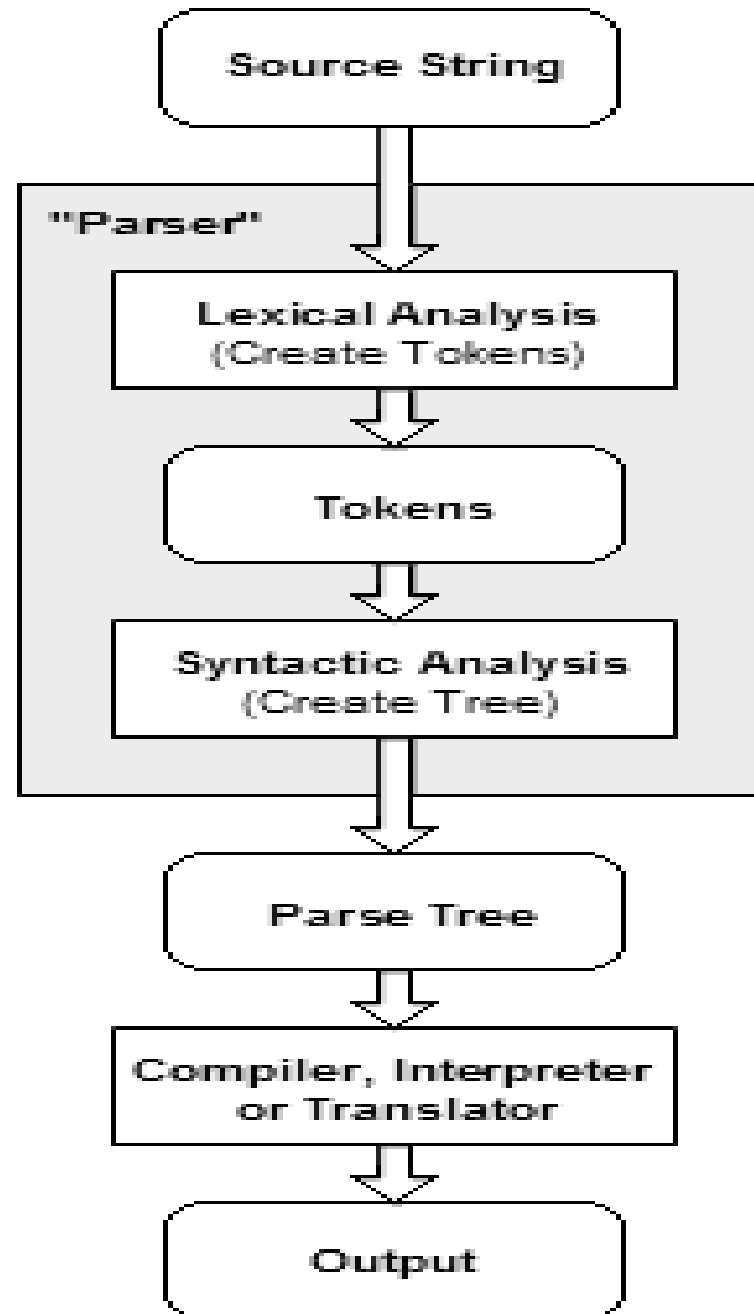
Bottom-up parsing

- Bottom-up parsing builds a parse tree from the leaves (terminals) to the start symbol

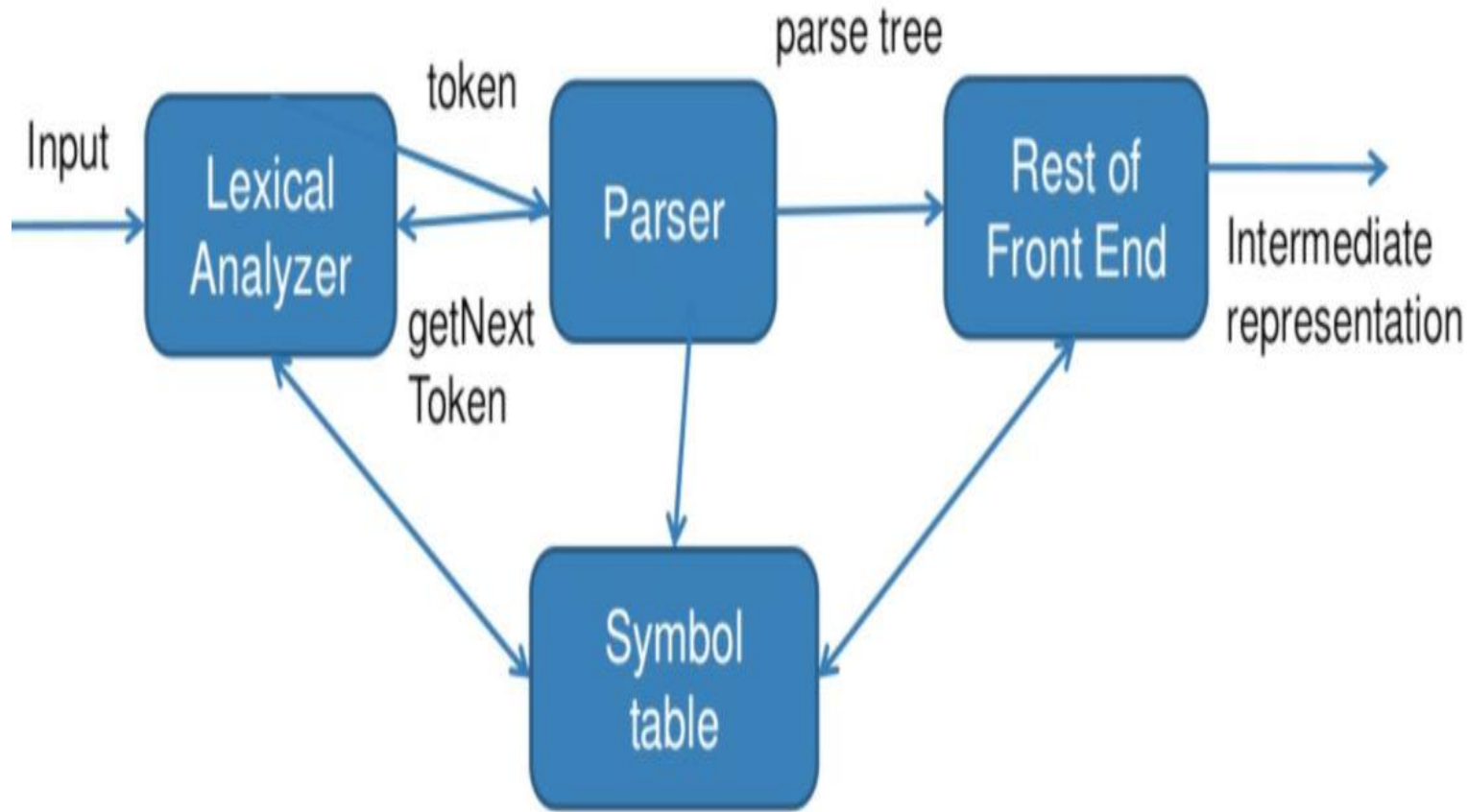








Add pos tagger in-between

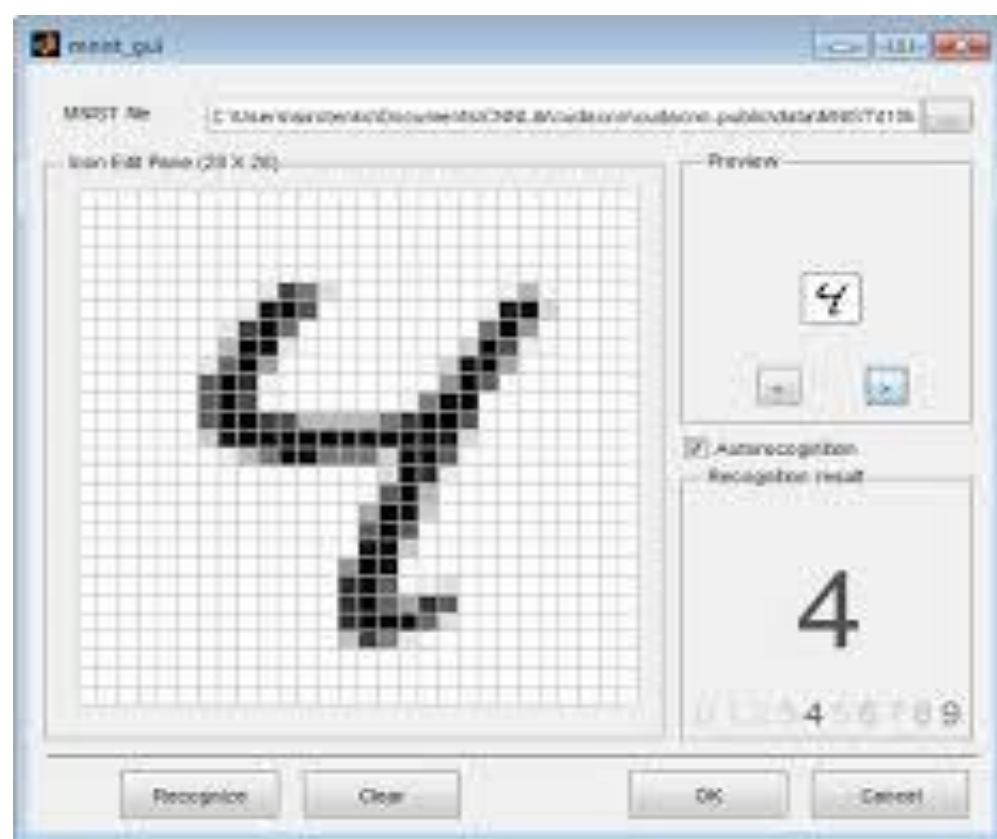


The **lexical analysis in NLP** deals with the study at the level of words with respect to their **lexical** meaning and part-of-speech. ... A lexeme is a basic unit of **lexical** meaning; which is an abstract unit of morphological **analysis** that represents the set of forms or “senses” taken by a single morpheme

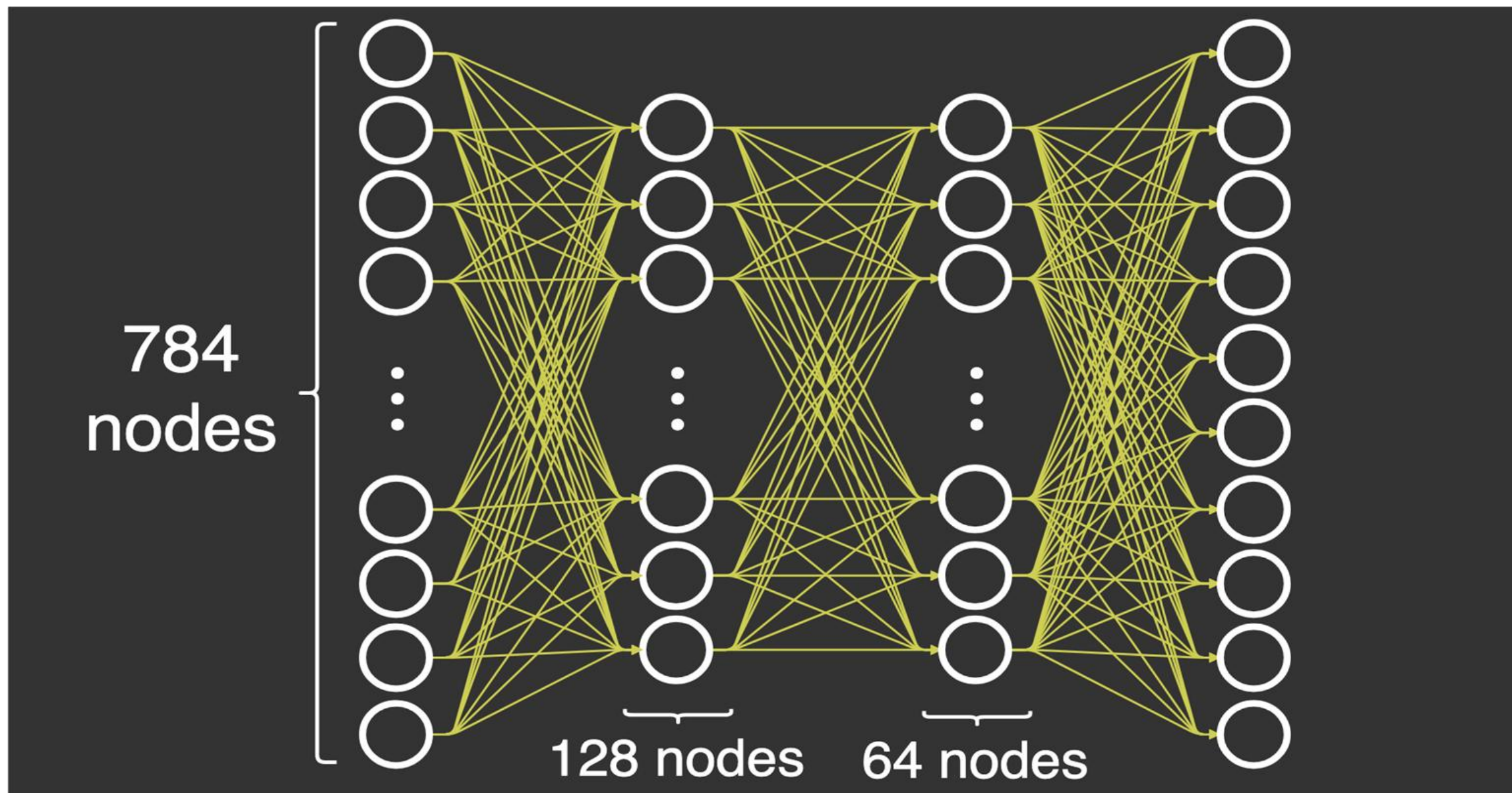
Optical character recognition



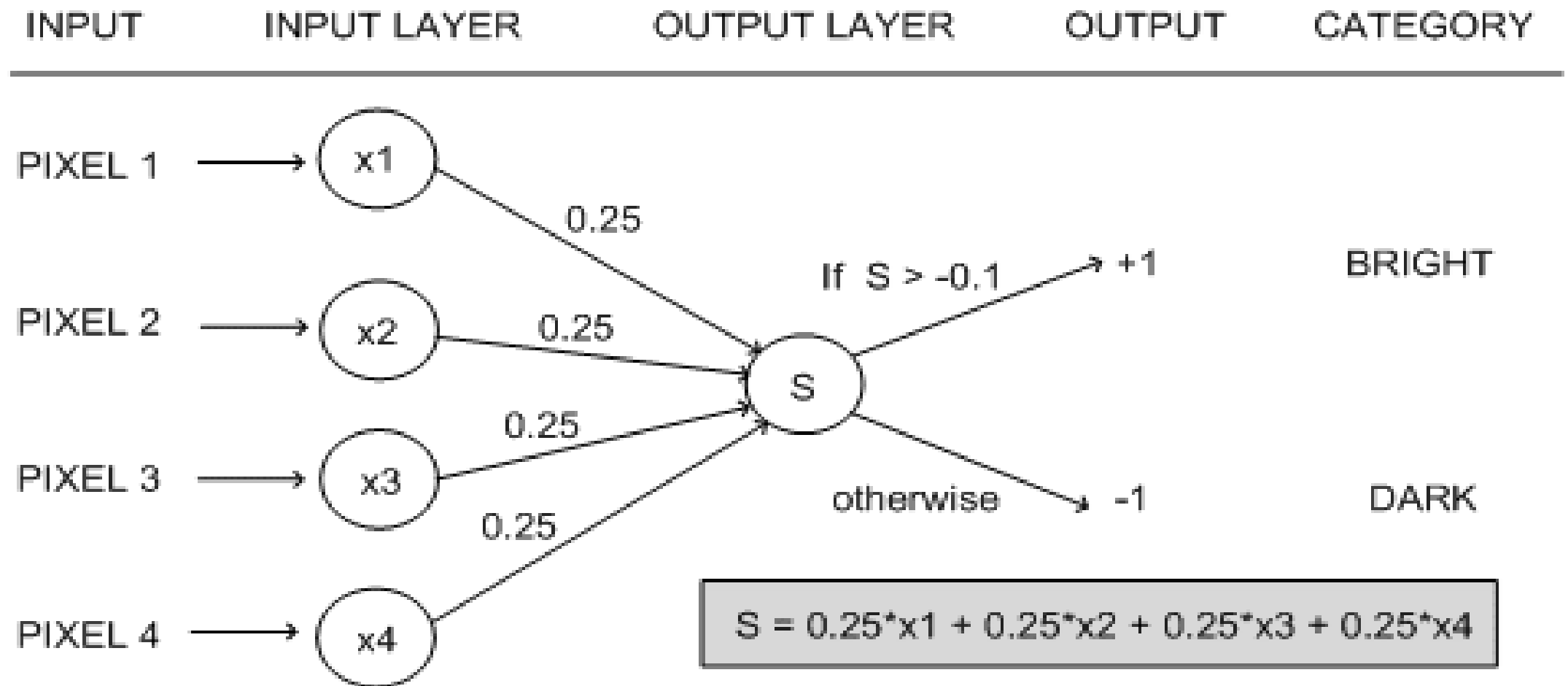
Hand writing recognizer



[illegible]



Michael Nilsen deep learning and
neural networks



$$F(x) = w \cdot x + b > 0 \quad (x \text{ is between } 0 \text{ and } 1)$$

$$\sum w_1 x_1$$

Local minima and maxima , cost function

	→	0.104376756903 letter a 0.0721450504693 letter d 0.0478821721917 letter m
	→	0.964058625761 letter m 0.428703615318 letter n 0.0400231174402 letter q
	→	0.999791090194 letter j 0.200741266115 letter s 0.00938691743516 letter y
	→	0.658396987672 letter s 0.0173122282696 letter k 0.00193613318379 letter z
	→	0.996591459276 letter z 0.00747525050305 letter j 0.00380705675129 letter c